Flipkart Logistics Optimization - SQL Queries

Data Base Creating :

```
CREATE DATABASE flipkart_logistics;
```

Creating tables :

Orders table creating:

```
CREATE TABLE Orders (
    Order_ID VARCHAR(20) PRIMARY KEY,
    Warehouse_ID VARCHAR(20),
    Route_ID VARCHAR(20),
    Agent_ID VARCHAR(20),
    Order_Date DATE,
    Actual_Delivery_Date DATE,
    Delivery_Status VARCHAR(50),
    Order_Value DECIMAL(10,2),
    Traffic_Delay_Min INT
);
```

Creating table Routes :

```
CREATE TABLE Routes (
    Route_ID VARCHAR(20) PRIMARY KEY,
    Start_Location VARCHAR(100),
```

```
    End_Location VARCHAR(100),
    Distance_KM DECIMAL(6,2),
    Avg_Travel_Time_Min INT,
    Traffic_Delay_Min INT
);
```

Creating table Warehouses :

```
CREATE TABLE Warehouses (
    Warehouse_ID VARCHAR(20) PRIMARY KEY,
    Warehouse_Name VARCHAR(100),
    City VARCHAR(100),
    Processing_Capacity INT,
    Avg_Processing_Time INT
);
```

Creating table Delivery Agents:

```
CREATE TABLE Delivery_Agents (
    Agent_ID VARCHAR(20) PRIMARY KEY,
    Agent_Name VARCHAR(100),
    Route_ID VARCHAR(20),
    Avg_Speed_KM_H DECIMAL(5,2),
    Delivery_Efficiency DECIMAL(5,2),
    Experience_Years INT
);
```

Creating table Shipment Tracking:

```
CREATE TABLE Shipment_Tracking (
```

```
    Shipment_ID VARCHAR(20) PRIMARY KEY,
    Order_ID VARCHAR(20),
    Route_ID VARCHAR(20),
    Checkpoint VARCHAR(100),
    Checkpoint_Time DATETIME,
    Delay_Reason VARCHAR(100)
);
```

Task 1 – Data Cleaning

Remove duplicate Order_ID

```
SELECT Order_ID, COUNT(*)
FROM Orders
GROUP BY Order_ID
HAVING COUNT(*) > 1;
```

Replace NULL Traffic Delay

```
UPDATE Orders o
JOIN (
    SELECT Route_ID, AVG(Traffic_Delay_Min) AS avg_delay
    FROM Orders
    GROUP BY Route_ID
) r
ON o.Route_ID = r.Route_ID
SET o.Traffic_Delay_Min = r.avg_delay
WHERE o.Traffic_Delay_Min IS NULL;
```

Invalid date check

```
SELECT *
FROM Orders
WHERE Actual_Delivery_Date < Order_Date;
```

Task 2 – Delay Analysis

Delay per order

```
SELECT Order_ID,
DATEDIFF(Actual_Delivery_Date, Order_Date) AS delay_days
FROM Orders;
```

Average delay per route

```
SELECT Route_ID,
AVG(DATEDIFF(Actual_Delivery_Date, Order_Date)) AS avg_delay
FROM Orders
GROUP BY Route_ID;
```

-Rank delays per warehouse

```
SELECT Order_ID, Warehouse_ID,
DATEDIFF(Actual_Delivery_Date, Order_Date) AS delay_days,
RANK() OVER (PARTITION BY Warehouse_ID ORDER BY delay_days DESC) AS
rank_delay
FROM Orders;
```

Task 3 – Route Optimization

Average delivery time per route

```
SELECT Route_ID,
AVG(DATEDIFF(Actual_Delivery_Date, Order_Date)) AS avg_time
FROM Orders
GROUP BY Route_ID;
```

Efficiency ratio

```
SELECT Route_ID,
(Distance_KM / Avg_Travel_Time_Min) AS efficiency
FROM Routes;
```

Worst routes

```
SELECT Route_ID,
(Distance_KM / Avg_Travel_Time_Min) AS efficiency
FROM Routes
ORDER BY efficiency ASC
LIMIT 3;
```

Task 4 – Warehouse Performance

```
SELECT Warehouse_ID,
COUNT(*) AS total_orders,
SUM(CASE
   WHEN DATEDIFF(Actual_Delivery_Date, Order_Date) > 0 THEN 1
   ELSE 0 END) AS delayed,
(SUM(CASE
   WHEN DATEDIFF(Actual_Delivery_Date, Order_Date) > 0 THEN 1
   ELSE 0 END) * 100.0 / COUNT(*)) AS delay_percentage
```

```
FROM Orders
GROUP BY Warehouse_ID;
```

Task 5 – Agent Performance

```
SELECT Agent_ID,
(SUM(CASE
    WHEN Delivery_Status = 'Delivered' THEN 1
    ELSE 0 END) * 100.0 / COUNT(*)) AS on_time_percentage
FROM Orders
GROUP BY Agent_ID;
```

Task 6 – Shipment Tracking

Latest checkpoint

```
SELECT t.*
FROM shipment_tracking t
JOIN (
    SELECT order_id, MAX(checkpoint_time) AS max_time
    FROM shipment_tracking
    GROUP BY order_id
) latest
ON t.order_id = latest.order_id
AND t.checkpoint_time = latest.max_time;
```

Delay reasons

```
SELECT Delay_Reason, COUNT(*)
FROM shipment_tracking
WHERE Delay_Reason <> 'None'
```

```
GROUP BY Delay_Reason;
```

Task 7 – KPI Queries

On-Time Delivery %

```
SELECT
(SUM(CASE
    WHEN Delivery_Status = 'Delivered' THEN 1
    ELSE 0 END) * 100.0 / COUNT(*)) AS on_time_percentage
FROM Orders;
```

Avg Traffic Delay per Route

```
SELECT Route_ID,
AVG(Traffic_Delay_Min)
FROM Routes
GROUP BY Route_ID;
```